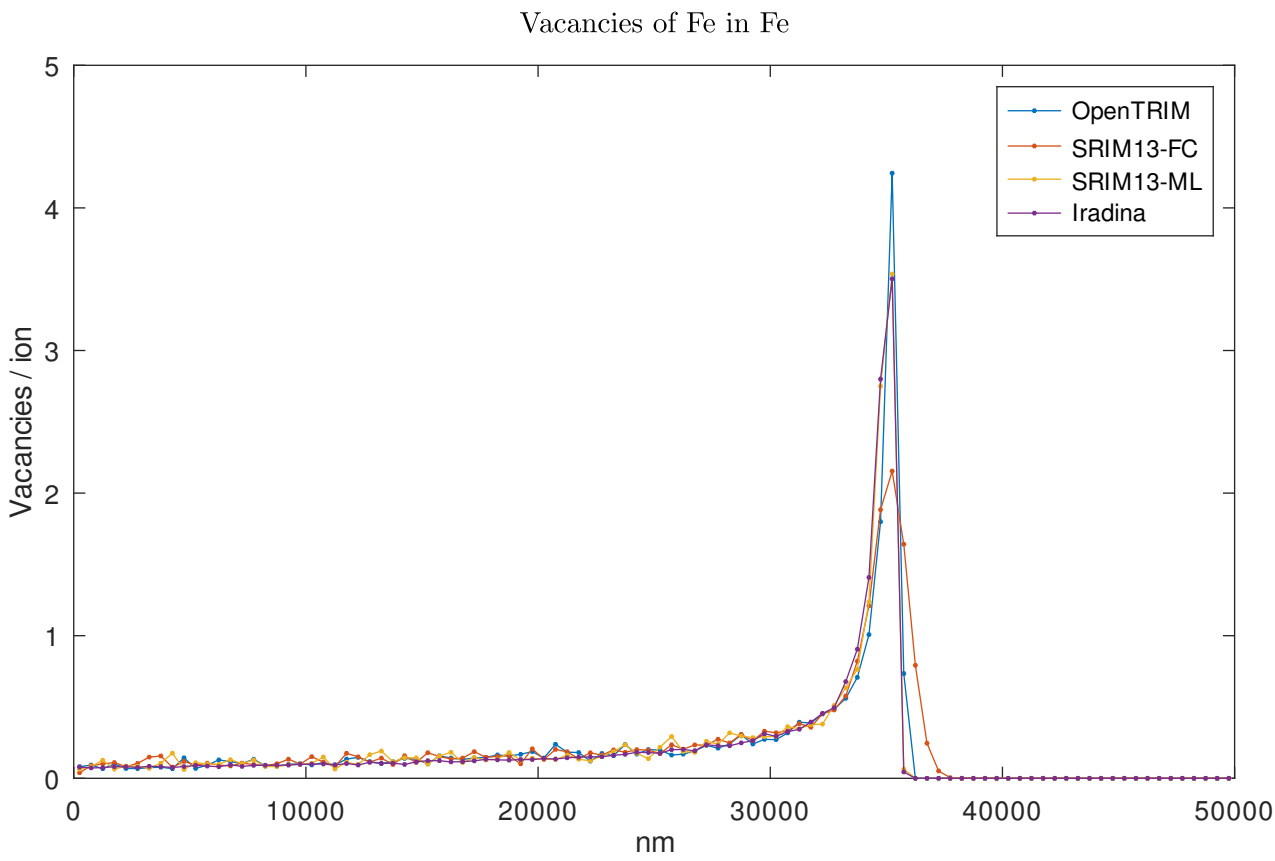
Benchmark #6
3 MeV H on Fe

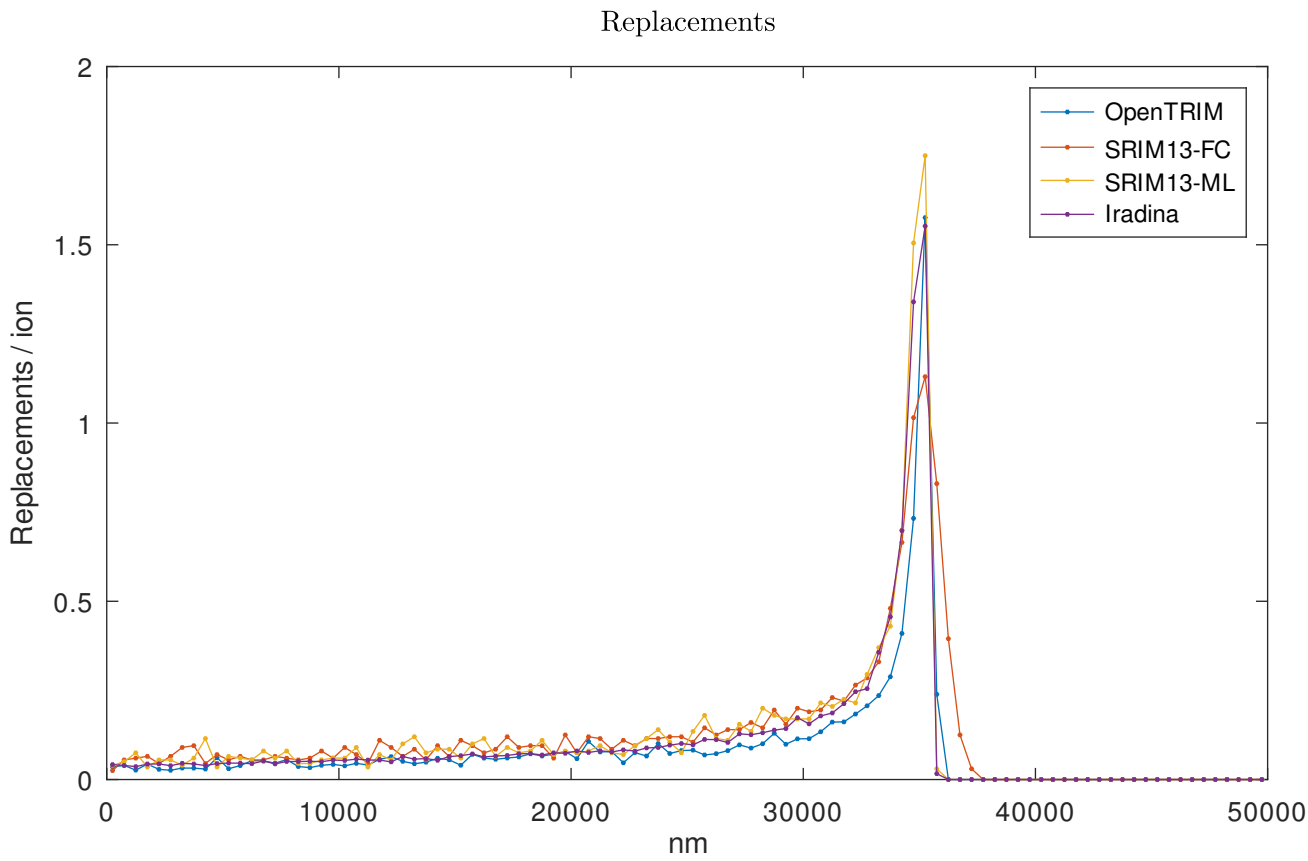
Ion energy E0 = 3e+06 eV

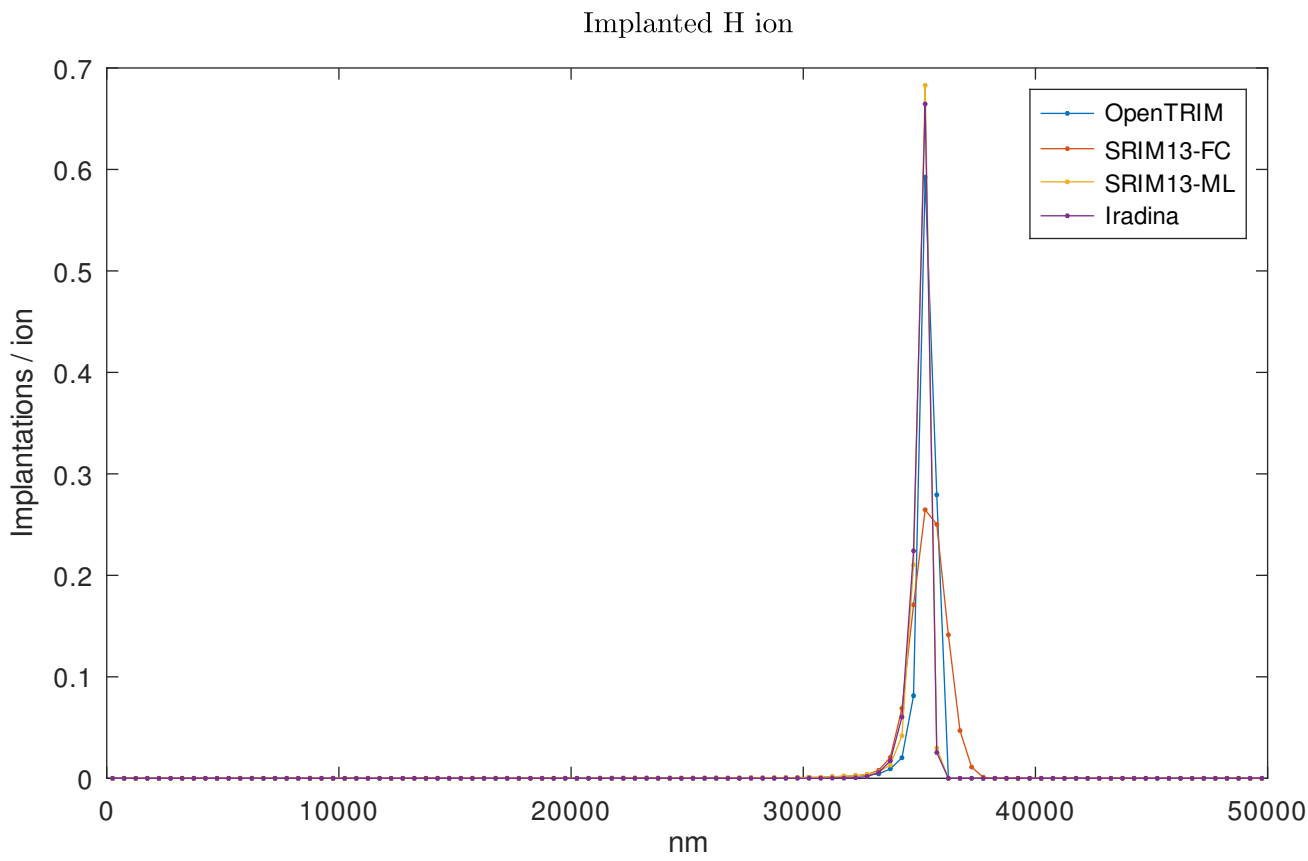
Target depth = 50000 nm

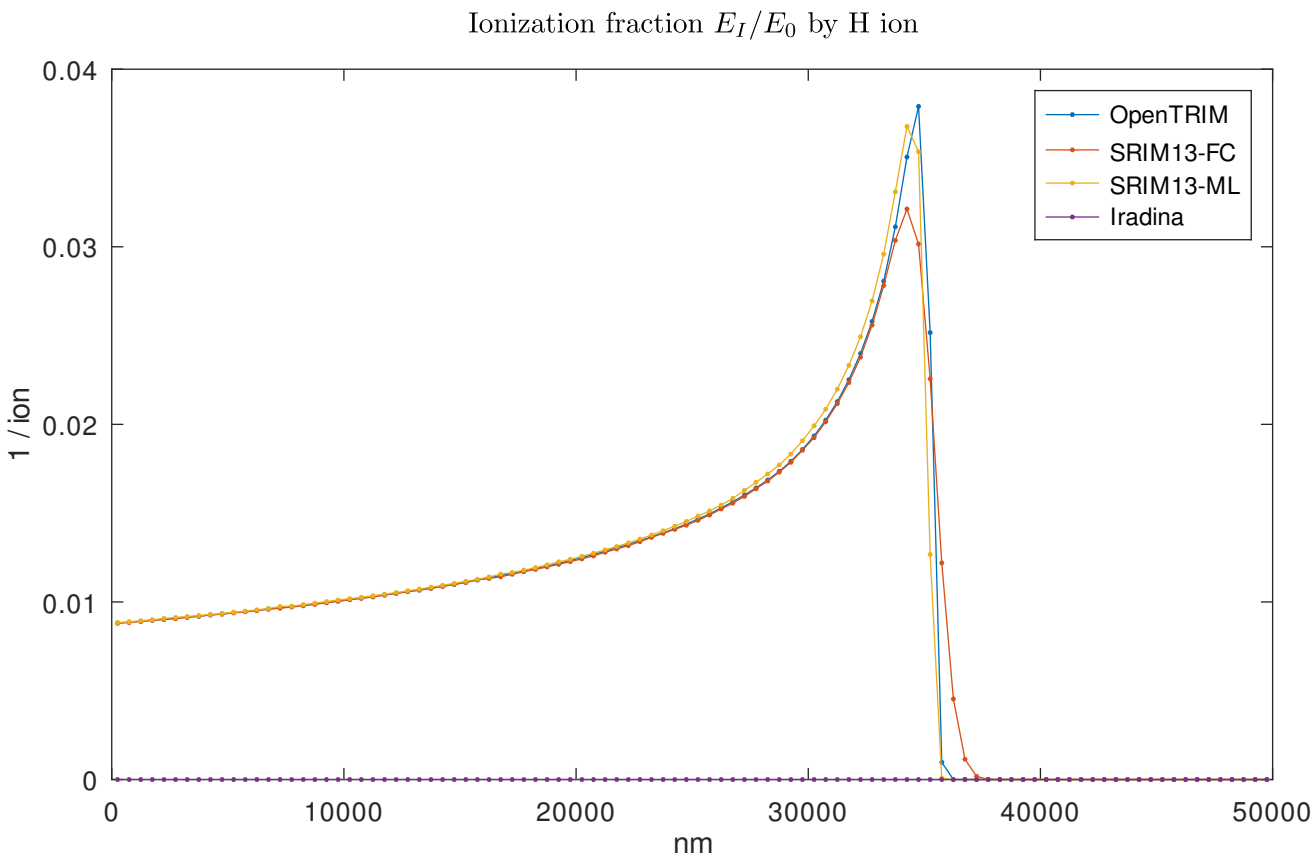
Summary Table

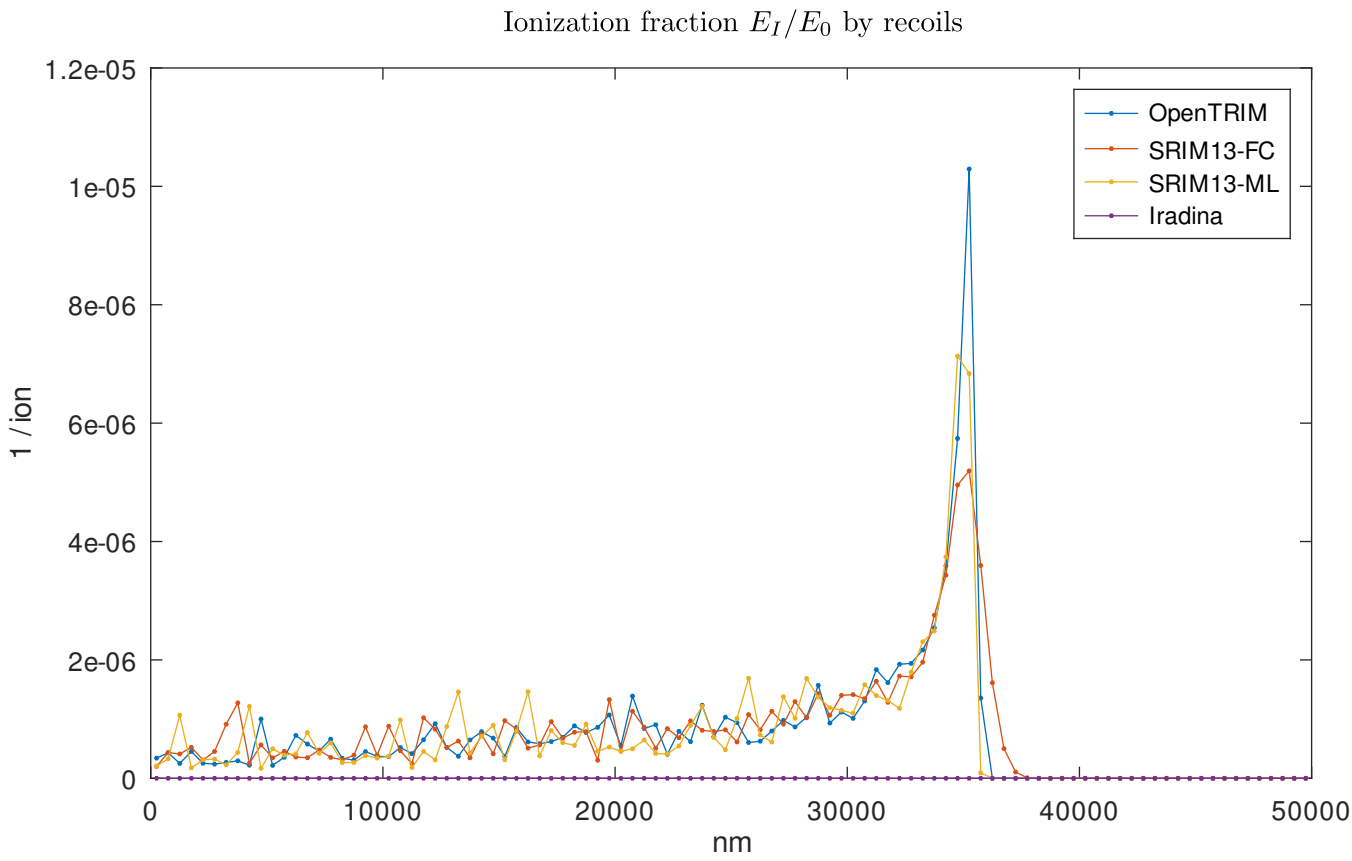
Quantity	OpenTRIM	SRIM13-FC	SRIM13-ML	Iradina
V(Fe)	19.9	20.9	20	19.5
R(tot)	8.07	11.9	11.4	9.95
I(H)	1	1	1	1
EI(H)/E0	0.999	0.999	0.999	0
EI(r)/E0	7.47e-05	7.53e-05	7.14e-05	0
EI/E0	0.999	0.999	0.999	0.999
EPh(H)/E0	0.000364	0.000316	0.000486	0
EPh(r)/E0	0.000539	0.000848	0.000994	0
EPh(tot)/E0	0.000903	0.00116	0.00148	0.000976
1-(EI+EPh)/E0	0	-0.000302	-0.000382	-4.77e-07

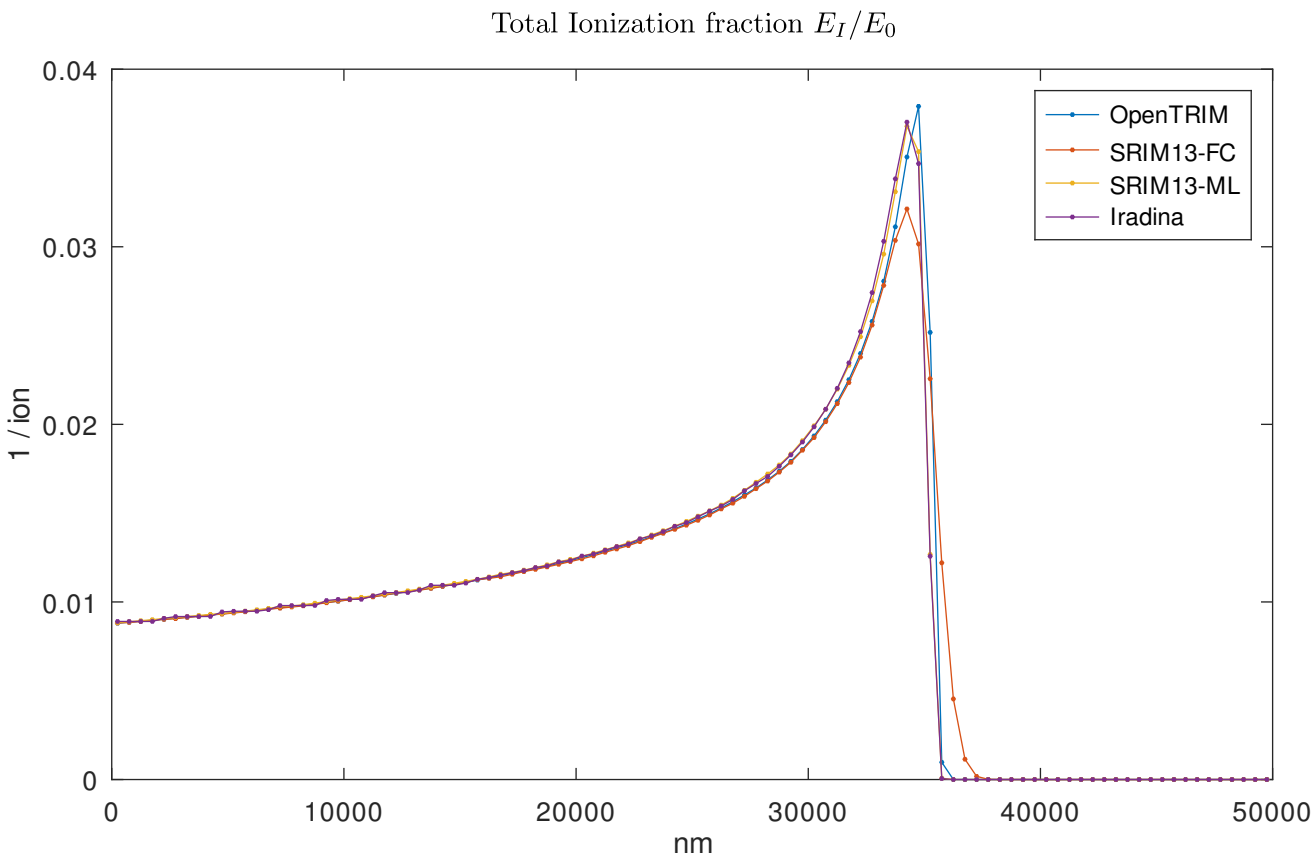

Vacancies of Fe in Fe
Vacancies / ion
nm
0
1
2
3
4
5
0
10000
20000
30000
40000
50000
OpenTRIM
SRIM13-FC
SRIM13-ML
Iradina

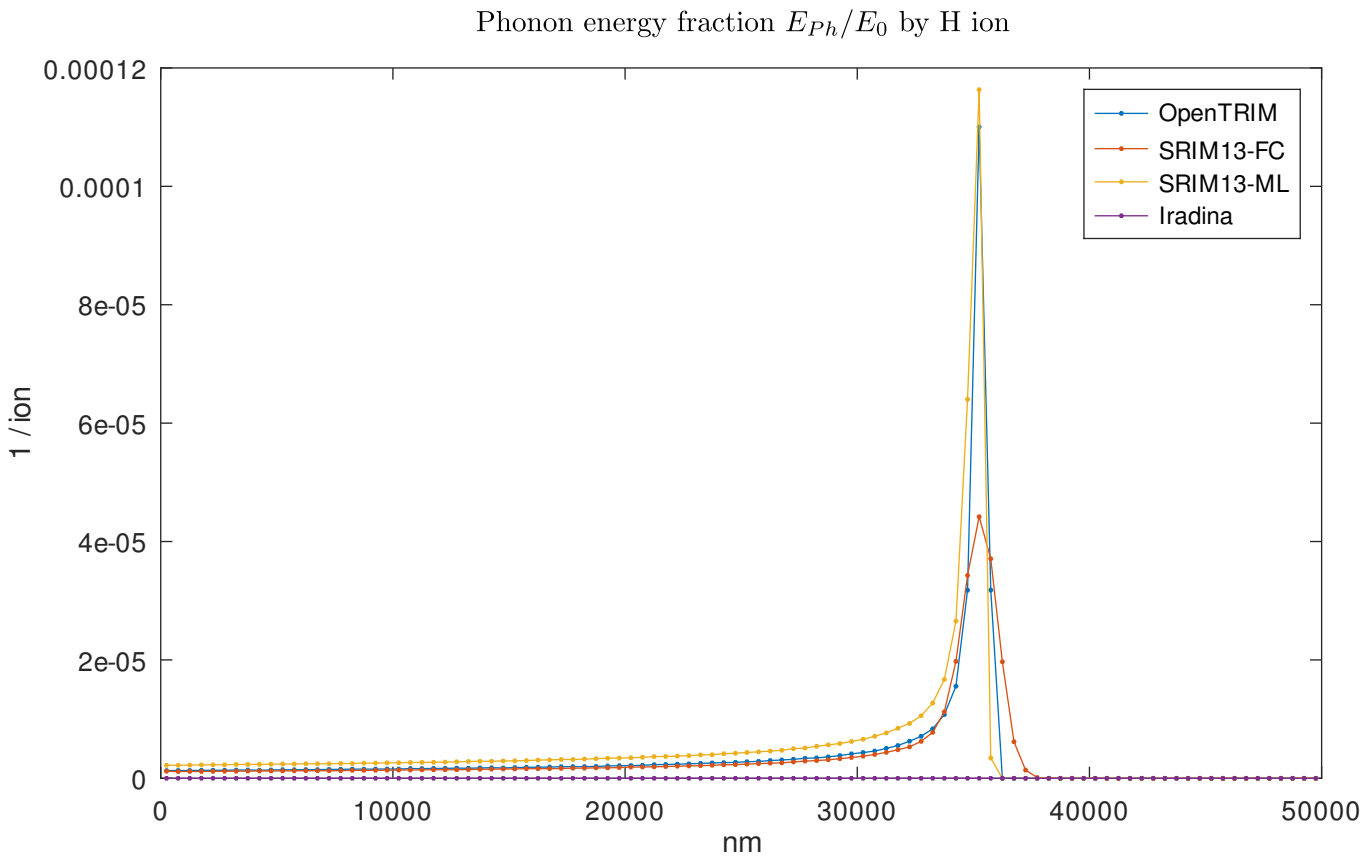

Replacements
Replacements / ion
nm
0
0.5
1
1.5
2
0
10000
20000
30000
40000
50000
OpenTRIM
SRIM13-FC
SRIM13-ML
Iradina

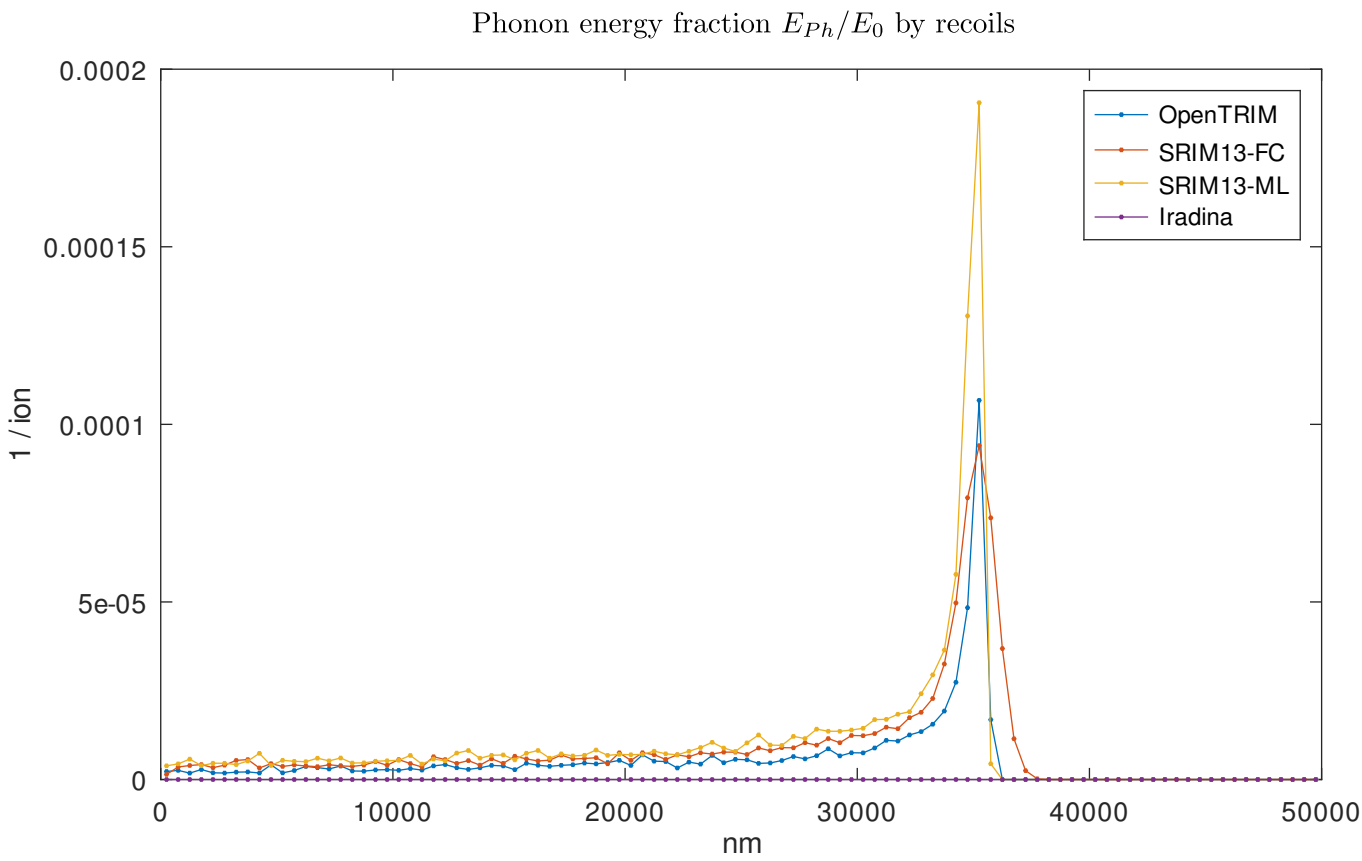

Implanted H ion
Implantations / ion
nm
OpenTRIM
SRIM13-FC
SRIM13-ML
Iradina
0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0
10000
20000
30000
40000
50000

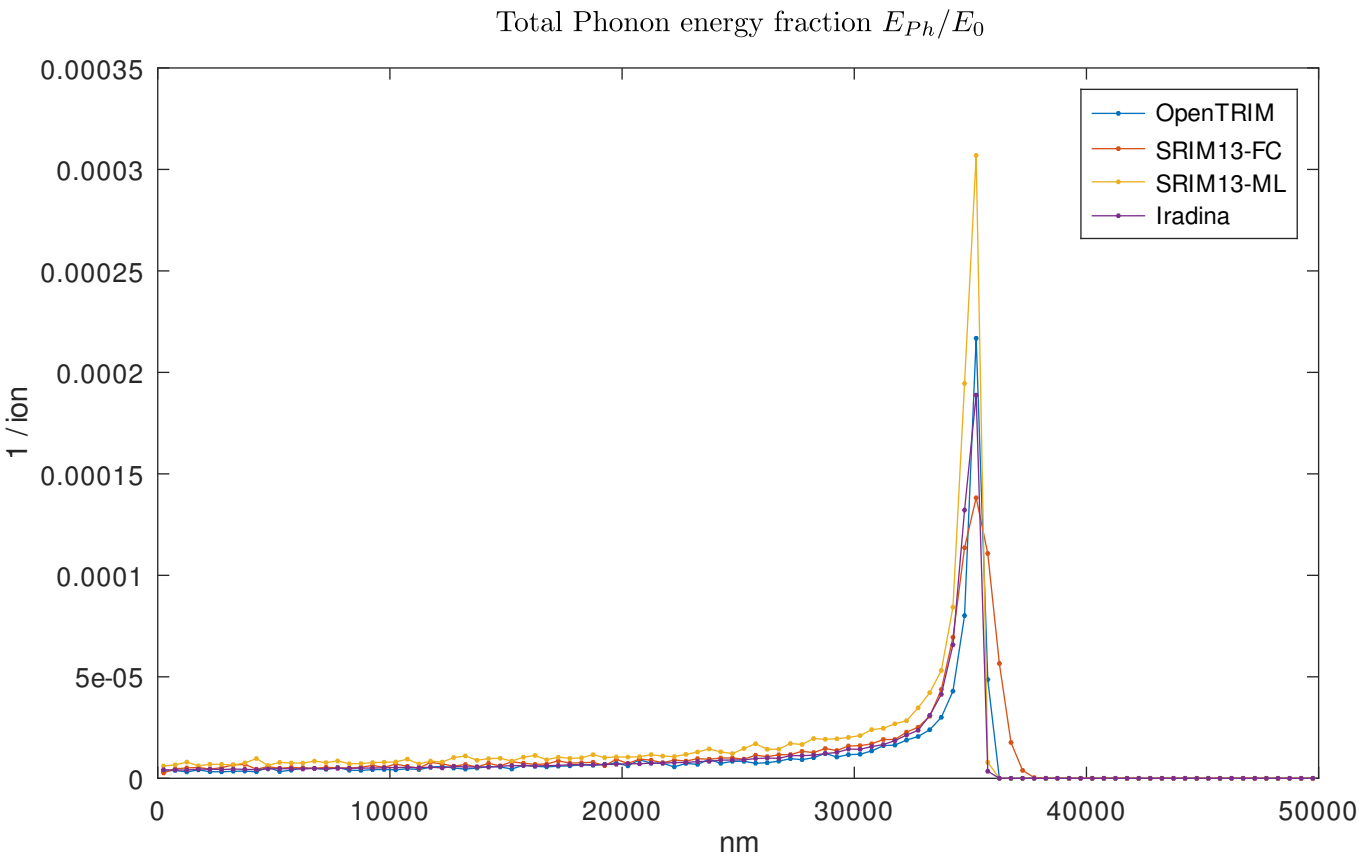

Ionization fraction E_I/E_0 by H ion
1 / ion
nm
OpenTRIM
SRIM13-FC
SRIM13-ML
Iradina
0
0.01
0.02
0.03
0.04
0
10000
20000
30000
40000
50000

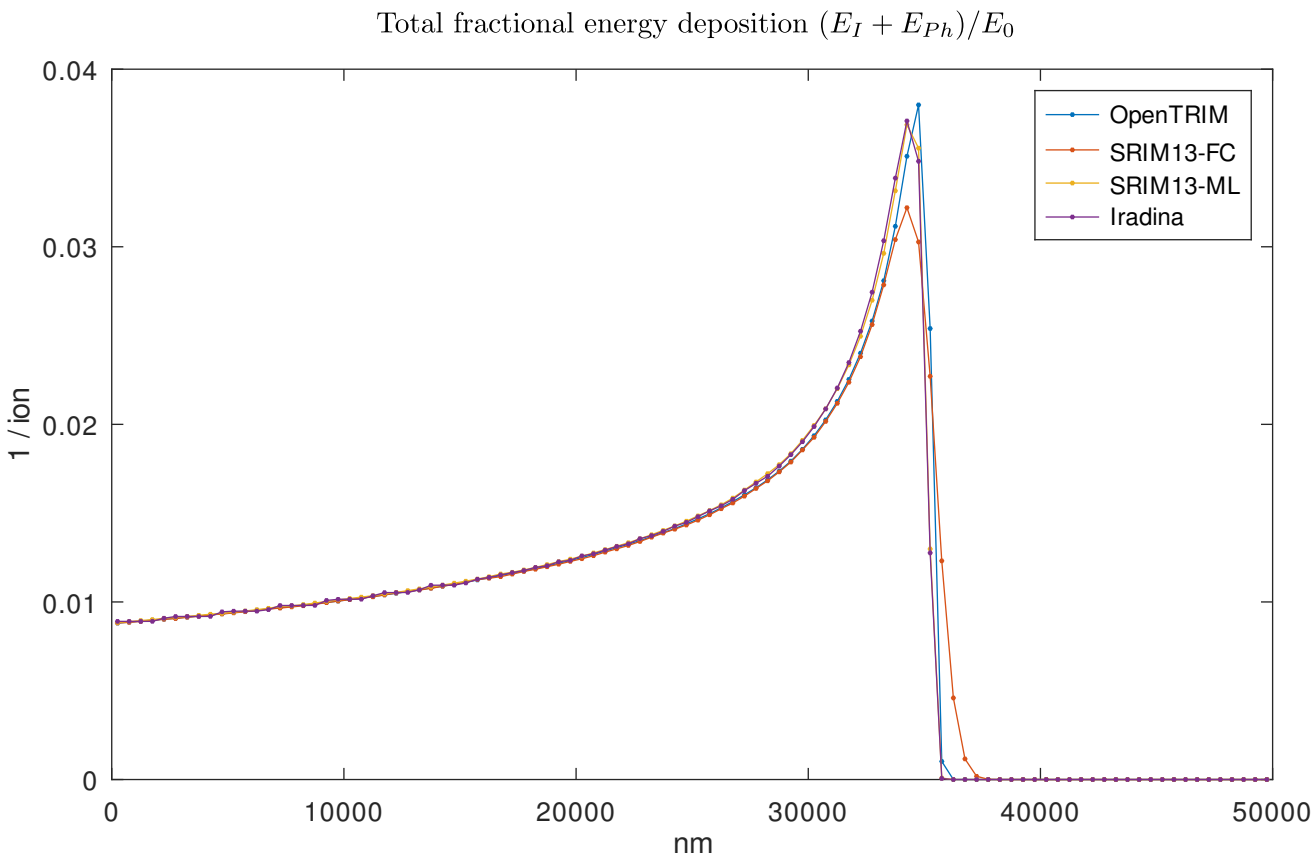

Ionization fraction E_I/E_0 by recoils
1.2e-05
1e-05
8e-06
6e-06
4e-06
2e-06
0
1 / ion
0
10000
20000
30000
40000
50000
nm
OpenTRIM
SRIM13-FC
SRIM13-ML
Iradina


Total Ionization fraction E_I/E_0
0.04
0.03
0.02
0.01
0
1 / ion
0
10000
20000
30000
40000
50000
nm
OpenTRIM
SRIM13-FC
SRIM13-ML
Iradina


Phonon energy fraction E_{Ph}/E_0 by H ion
0.00012
0.0001
8e-05
6e-05
4e-05
2e-05
0
1 / ion
0
10000
20000
30000
40000
50000
nm
OpenTRIM
SRIM13-FC
SRIM13-ML
Iradina


Phonon energy fraction E_{Ph}/E_0 by recoils
0.0002
0.00015
0.0001
5e-05
0
1 / ion
0
10000
20000
30000
40000
50000
nm
OpenTRIM
SRIM13-FC
SRIM13-ML
Iradina


Total Phonon energy fraction E_{Ph}/E_0
0.00035
0.0003
0.00025
0.0002
0.00015
0.0001
5e-05
0
1 / ion
0
10000
20000
30000
40000
50000
nm
OpenTRIM
SRIM13-FC
SRIM13-ML
Iradina


Total fractional energy deposition $(E_I + E_{Ph})/E_0$
0.04
0.03
0.02
0.01
0
1 / ion
0
10000
20000
30000
40000
50000
nm
OpenTRIM
SRIM13-FC
SRIM13-ML
Iradina